BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 1: Calculations and Extended Answers

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is a constant?

2. What does an IDE's run-time environment let you do?

A) Run and test code
B) Repair physical CPU faults
C) Rearrange HDD fragments
D) Manage network user accounts

3. Where are the MSB and LSB in an eight-bit unsigned number?

4. A file stores text, not a photo of a page. "Every letter A is stored as a tiny bitmap." Correct this. How is a letter stored and read?

Today's objectives

Explain the core ideas of paper 1: calculations and extended answers using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Show formulas
Recall and write the calculation rule before substitutions.

Keep units consistent
Convert bits/bytes and MB/GB before arithmetic.

Answer command words
Describe what happens, explain why and evaluate trade-offs.

Earn full tariff
Add linked causal explanations, not repeated versions of one fact.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Show formulas	Recall and write the calculation rule before substitutions.
Keep units consistent	Convert bits/bytes and MB/GB before arithmetic.
Answer command words	Describe what happens, explain why and evaluate trade-offs.
Earn full tariff	Add linked causal explanations, not repeated versions of one fact.

I Do - One worked example

Calculate image size: 120 x 80 pixels at 4-bit colour depth, in bytes.

1. Pixels: 120 x 80 = 9600.
2. Size: 9600 x 4=38,400 bits.
3. Divide eight = 4800 bytes.
4. Identify all figures and correct units.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Calculate sound file bits: 8000 Hz, 5 seconds, 8 bits/sample (mono).

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change the bit depth and predict the new uncompressed raw data size in bytes.

```
width = 12
height = 10
bitsPerPixel = 4
bytes = width * height * bitsPerPixel DIV 8
print(bytes)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Calculate 2560 bits in bytes. [2 marks]

Q3. Convert 3 MB to kB under decimal units. [2 marks]

Q4. Explain why two distinct SSD benefits may score better than repeating 'fast'. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. When calculating sound size, sampling rate can be ignored. ☐ T ☐ F
2. Recall and write the calculation rule before substitutions. ☐ T ☐ F
3. Convert bits/bytes and MB/GB before arithmetic. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 A named value that stays the same while the program runs. ☐

Q2 A - Run and test code ☐

Q3 The MSB is the leftmost bit, weight 128; the LSB is the rightmost bit, weight 1. ☐

Q4 Each character has a number code stored in binary. Both computers need a matching encoding to show the intended characters. ☐

Code First - Model

```
width = 12
height = 10
bitsPerPixel = 8
bytes = width * height * bitsPerPixel DIV 8
print(bytes)
```

Output: 120

We Do - Check the model

1. No header/compression assumptions. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 2560 / 8 = 320 bytes. ☐

Q3 3000 kB. ☐

Q4 Specific benefits (e.g. impact durability and access time) address different marking points. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: